

Lirui Wang

5000 25th Ave NE, Seattle, WA 98105

✉ liruiw@uw.edu

🌐 liruiw.github.io

☎ +1 (206) 483-8515

EDUCATION

University of Washington, Seattle, WA

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

Sep. 2016 – Jun. 2020

GPA: 3.97 / 4.00, Advisor: Prof. Dieter Fox

University of Washington, Seattle, WA

Master of Science, Computer Science, College of Arts and Sciences

Sep. 2020 – Jun. 2021

RESEARCH PROJECTS

Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection, Seattle, WA

UW Robotics and States Estimation Lab

March. 2019 – Present.

- Implement a network that combines data-driven and traditional method to synthesize grasp pose.
- Prototype RL, IL, and MPC with PyBullet environment to map observed points directly to 6D pose increments in task space.
- Introduce an integrate planner that models joint motion and grasp planning as an optimization problem.
- Propose an online learning framework and a grasp synthesis approach to generate grasp and select grasp online.

6D Object Pose Tracking with the Deep Iterative Matching Network, Seattle, WA

UW Robotics and States Estimation Lab

June. 2018 – March. 2019

- Tested DeepIM on YCB-Video datasets and achieved robust results trained on synthetic data only
- Improved network training stability, pose distribution and occlusion generation
- Extended DeepIM in real-time tracking for occluded scene in YCB-Video dataset and built AR demos
- Built renderer and kinematics tree for vision system application on robotics.

Unsupervised Flow and Depth Methods Study, Seattle, WA

UW Robotics and States Estimation Lab

Jan. 2018 – Jun. 2018

- Investigated a few deep learning methods for 3D geometry tasks such as depths, flow, and egomotion.
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting

Baxter Hand-Eye Coordination, Seattle, WA

UW Robotics and States Estimation Lab

Aug. 2017 – Jan. 2018

- Worked on a manipulation task with PoseCNN, which enables Baxter to move daily objects.
 - Studied on camera models, hand-eye calibration with depth camera, robot kinematics, 3D geometry and ROS packages for tracking and control.
-

EXPERIENCE

Teaching Assistant, Seattle, WA

UW CSE 455 (Computer Vision)

Sep. 2019 – Present.

Research Assistant, Seattle, WA

UW Adaptive Computing Machines and Emulator Lab

March. 2018 – Jun. 2018

- Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle

Computer Vision Lead & Drone Lead, Seattle, WA

Advanced Robotics at the University of Washington

Sep. 2016 – Feb. 2018

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI
- Perform system integration of controls, navigation and serial communication RoboMasters hosted by DJI

Robotics Engineer Intern, Shenzhen, China

DJI Technology

Jun. 2017 – Aug. 2017

- Worked on automatic robot battle of vehicles with TX series onboard and DJI Manifold with Guidance
 - Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
 - Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot
-

SKILLS & OTHERS

Programming Language: Java, C++/C, Python, Matlab

Frameworks: ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32

Tools: Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt

Courses: Computer Vision, Deep Learning, Artificial Intelligence, Graphics, Robotics, Algorithms, Numerical Analysis, Linear Analysis, Convex Analysis and Optimization, Embedded Systems, Control Theory, Digital Signal Processing, Statistical Learning, High-Dimension Statistics, Robust Machine Learning

Scholarships: Tau Beta Pi Scholarship 2017-2019, Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship 2017-2018, Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship 2018-2019